EXPERIMENT 9:

Analysis of Photodetector Noise and Signal-to-Noise Ratio (SNR) using MATLAB.

Objective :

To analyze shot noise, dark current noise, and thermal noise in a photodetector using MATLAB, calculate the signal-to-noise ratio (SNR), and determine the minimum detectable optical power.

SOFTWARE REQUIRED:

MATLAB.

MATLAB PROGRAM:

```
% Photodetector Noise and SNR Analysis

clc;

clear;

close all;


%% 1. Constants and Input Parameters

q = 1.602e-19;

kB = 1.381e-23;

h = 6.626e-34;

c = 2.998e8;


lambda = 1550e-9;

eta = 0.85;

Id = 5e-9;

T = 300;

RL = 50;
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Fn = 2;
B = 1e9;

Pdbm = -60:0.1:0;
P = 10.^((Pdbm-30)/10);

%% 2. Responsivity and Signal Current
R = eta*q*lambda/(h*c);
Ip = R*P;

%% 3. Noise Variances and SNR
var_shot = 2*q*Ip*B;
var_dark = 2*q*Id*B;
var_th = 4*kB*T*B*Fn/RL;

var_tot = var_shot + var_dark + var_th;
SNR_dB = 10*log10(Ip.^2 ./ var_tot);

%% 4. Minimum Detectable Power (SNR = 1)
a = R^2;
b = -2*q*B*R;
cc = -(var_dark + var_th);

Pmin = (-b + sqrt(b^2 - 4*a*cc))/(2*a);
PmindBm = 10*log10(Pmin) + 30;

%% 5. Display Main Results
k = find(abs(Pdbm + 30) < 1e-9, 1);
```

```
fprintf('Responsivity R = %.4f A/W\n', R);
fprintf('Thermal variance = %.3e A^2\n', var_th);
fprintf('Dark variance = %.3e A^2\n', var_dark);
fprintf('Shot variance (-30 dBm) = %.3e A^2\n', var_shot(k));
fprintf('SNR at -30 dBm = %.2f dB\n', SNR_dB(k));
fprintf('Pmin (SNR = 1) = %.2f dBm (%.3f nW)\n\n', ...
    PmindBm, Pmin*1e9);

%% 6. Test Cases: -50 dBm, -30 dBm and 0 dBm
test_dBm = [-50 -30 0];

for i = 1:length(test_dBm)

    Pin_dBm = test_dBm(i);
    Pin = 10^((Pin_dBm-30)/10);

    I_signal = R*Pin;
    shot = 2*q*I_signal*B;
    total = shot + var_dark + var_th;
    SNR = 10*log10(I_signal^2/total);

    fprintf('TEST CASE: POWER (%d dBm)\n', Pin_dBm);
    fprintf('Input power = %d dBm\n', Pin_dBm);
    fprintf('Signal current = %.3e A\n', I_signal);
    fprintf('Shot variance = %.3e A^2\n', shot);
    fprintf('Thermal variance = %.3e A^2\n', var_th);
    fprintf('Dark variance = %.3e A^2\n', var_dark);
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    fprintf('Total variance = %.3e A^2\n', total);
    fprintf('SNR = %.2f dB\n', SNR);

    if SNR < 0
        fprintf('Signal is below the 0 dB SNR level.\n\n');
    else
        fprintf('Signal is at or above the 0 dB SNR level.\n\n');
    end
end

%% 7. Graph 1: Noise Variance vs Input Power
figure;

loglog(P, var_th*ones(size(P)), 'LineWidth', 1.5);
hold on;
loglog(P, var_dark*ones(size(P)), 'LineWidth', 1.5);
loglog(P, var_shot, 'LineWidth', 1.5);
loglog(P, var_tot, 'LineWidth', 2);

grid on;
xlabel('Input Power (W)');
ylabel('Noise Variance (A^2)');
title('Noise Variance vs Input Power');

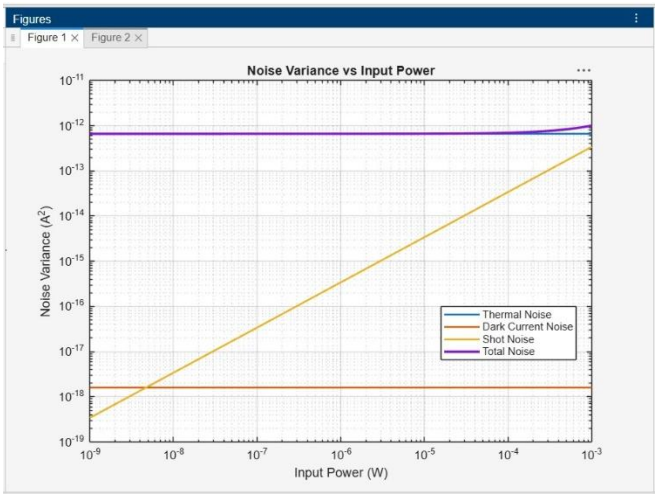
legend('Thermal Noise', 'Dark Current Noise', ...
       'Shot Noise', 'Total Noise', ...
       'Location', 'best');
```

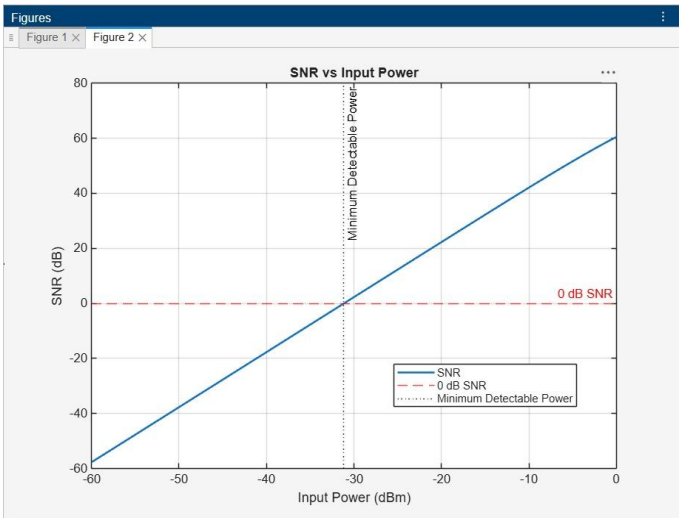
```
%% 8. Graph 2: SNR vs Input Power
figure;

plot(Pdbm, SNR_dB, 'LineWidth', 1.5);
hold on;
yline(0, 'r--', '0 dB SNR');
xline(PmindBm, 'k:', 'Minimum Detectable Power');

grid on;
xlabel('Input Power (dBm)');
ylabel('SNR (dB)');
title('SNR vs Input Power');
legend('SNR', '0 dB SNR', 'Minimum Detectable Power', ...
    'Location', 'best');
```

OUTPUTS:





REFLECTION:

The experiment helped me understand how shot noise, dark current noise and thermal noise affect photodetector performance. The MATLAB graphs show the relationship between input optical power and SNR.

. RUBRICS:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improve-ment – 1	Weight-age
Develop the MATLAB program with correct constants and system parameters.	Program Setup and Parameter Definition	Defines all constants and parameters correctly with proper units and a clean script structure, independently.	Defines parameters correctly with minor unit or syntax errors and limited assistance.	Defines most parameters but needs considerable assistance or has unit errors (e.g., dBm/W, linear F_n).	Unable to set up the script or define the parameters.	25
Compute R, signal current, noise variances and SNR across the power sweep.	Simulation Execution and Data Acquisition	Computes R, I_P, all noise variances and SNR correctly over −60 to 0 dBm and records the values.	Computes all quantities with minor errors or omissions.	Computes some quantities with noticeable errors or inconsistencies.	Unable to run the simulation or record the values.	25
Obtain the noise and SNR plots and determine P_{min}.	Graph Plotting and P_{min} Determination	Plots accurate log–log and semi-log curves with labels and reference lines; P_{min} within 5% of the reference.	Plots the curves with minor errors; P_{min} identified with small deviation.	Plots partial or poorly scaled curves; P_{min} approximate.	Unable to plot the curves or determine P_{min}.	25
Interpret noise regimes, SNR slopes and receiver sensitivity.	Analysis and Engineering Interpretation	Correctly identifies thermal- and shot-limited regimes, explains the SNR slopes and links results to receiver design.	Identifies the regimes with minor errors and reasonable interpretation.	Performs basic interpretation with noticeable errors or limited justification.	Unable to interpret the results.	25

CONCLUSION :

The photodetector noise and SNR were analyzed using MATLAB. The minimum detectable optical power was found to be approximately −31.16 dBm, and the noise and SNR characteristics were plotted successfully.